

The Big Idea S8P3

Motion = a change in an object's position over time. Forces (pushes and pulls) cause objects to start, stop, speed up, slow down, or change direction. We describe motion with **distance**, **displacement**, **speed**, **velocity**, and **acceleration**.

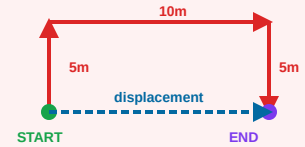
Distance vs. Displacement S8P3a

DISTANCE

TOTAL path length traveled. Only counts "how much."
Unit: meters (m). Always positive.

DISPLACEMENT

Straight-line change in position from start to end, *with direction*.
Unit: meters (m) + direction (N, S, E, W).



Distance = $5 + 10 + 5 = 20\text{m}$
Displacement = 10m right

Speed, Velocity & Acceleration S8P3a

SPEED

How fast something moves. No direction.

$$\text{speed} = \text{distance} \div \text{time}$$

Unit: m/s (meters per second)
Example: A car at 60 mph

VELOCITY

How fast AND which direction.

$$\text{velocity} = \text{displacement} \div \text{time}$$

Unit: m/s + direction
Example: 60 mph NORTH

ACCELERATION

Change in velocity over time. Speeding up, slowing down, OR changing direction!

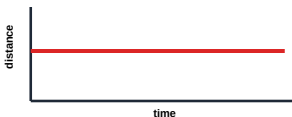
Unit: m/s^2
Example: car going from 0 to 60 mph

Key: SPEED just tells you how fast. VELOCITY tells you how fast AND where you're going. A car turning a corner at constant speed is STILL accelerating (direction is changing)!

Distance-Time Graphs S8P3a

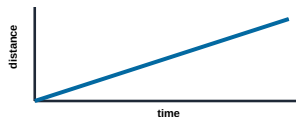
Shows how **distance (y-axis)** changes with **time (x-axis)**. The **SLOPE = SPEED**. Steeper = faster!

AT REST



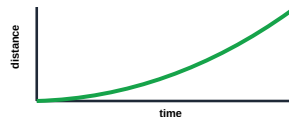
Flat line → no motion

CONSTANT SPEED



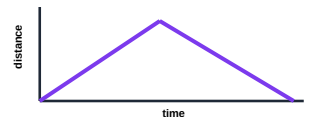
Straight slope → steady speed

ACCELERATING



Curve up → speeding up

RETURNING

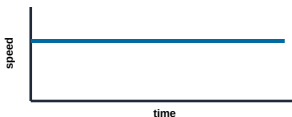


Slope down → heading back

Speed-Time (Velocity-Time) Graphs S8P3a

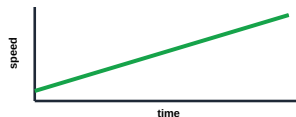
Shows how **speed (y-axis)** changes with **time (x-axis)**. The **SLOPE = ACCELERATION**.

CONSTANT SPEED



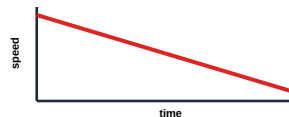
Flat → speed not changing ($a = 0$)

SPEEDING UP



Slope up → accelerating (+)

SLOWING DOWN



Slope down → decelerating (-)

AT REST



Line on x-axis → speed = 0

ACTIVITY 1: Motion & Graphs

- You walk 200 m to a store, then 200 m back home. Distance = ____ m. Displacement = ____ m.
- A runner goes 100 m EAST in 10 seconds. What is their velocity? ____ m/s ____.
- On a DISTANCE-TIME graph, a FLAT line means the object is ____.
- On a SPEED-TIME graph, a FLAT line means the object has ____ speed (acceleration = ____).
- A car turns left at a constant speed. Is it accelerating? ____ Why? ____

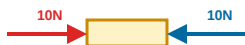
Flip over for forces, Newton's Laws, inertia, and free body diagrams →

Forces & Net Force S8P3b

A **force** is a push or a pull. Measured in **newtons (N)**. Forces have a size AND direction.

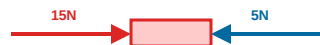
Net force = the sum of ALL forces acting on an object. It decides whether the object's motion will change!

BALANCED FORCES (net force = 0)



Object stays at rest OR keeps moving at constant velocity. NO change in motion.

UNBALANCED FORCES (net force ≠ 0)



Net force = 10 N right → object **ACCELERATES** in that direction.

Newton's Three Laws of Motion S8P3b

1 LAW OF INERTIA

An object at rest stays at rest. An object in motion stays in motion — UNLESS acted on by an unbalanced force.

Inertia = the tendency to resist a change in motion.

Real-world examples:

- Seatbelts stop YOU when the car stops suddenly
- A book sits on a table until pushed
- Tablecloth trick: pull fast, dishes stay!

2 $F = m \times a$

The force on an object equals its mass times its acceleration.

$$F = m \times a$$

More force → more acceleration. More mass → harder to accelerate.

Real-world examples:

- Pushing an empty grocery cart is easier than a full one
- Kicking a ball harder makes it go faster

3 ACTION-REACTION

For every action force, there is an **EQUAL** and **OPPOSITE** reaction force.

Forces come in **pairs**, acting on **TWO DIFFERENT** objects.

Real-world examples:

- You push off the ground → ground pushes you up (jumping!)
- Rocket fuel pushes down → rocket pushes up
- Swimmer pushes water back → water pushes swimmer forward

Inertia, Mass & Force S8P3c

More mass = more inertia = more force needed to change its motion. The amount of force needed to accelerate an object is proportional to its mass.



★ **Key claim:** For the SAME acceleration, a heavier object needs MORE force. For the SAME force, a heavier object accelerates LESS.

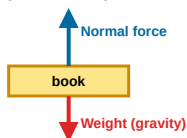
Evidence examples:

- A full shopping cart is harder to push than an empty one
- Adults need less force to stop a baby's stroller than a car
- It takes a lot of force to move a train, but very little to roll a basketball

Free Body Diagrams (FBDs)

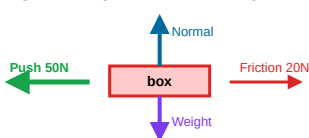
A **free body diagram** is a simple drawing showing ALL the forces acting ON an object as arrows. Arrow size = force size; arrow direction = force direction.

Book on table (balanced)



Equal arrows = **BALANCED** → no motion

Pushing a box (unbalanced →)



Net force = 50 - 20 = 30 N left → box **ACCELERATES**

Common Forces to Know

- **Weight/Gravity** — pulls DOWN
- **Normal** — surface pushes UP
- **Applied** — push or pull
- **Friction** — opposes motion
- **Tension** — pull of a rope

ACTIVITY 2: Newton's Laws

Write which law (1, 2, or 3) matches:

1. _____ A book stays still until someone picks it up
2. _____ A rocket pushes fuel down, which pushes the rocket up
3. _____ Pushing a heavy cart is harder than pushing a light one
4. _____ Passengers lurch forward when a bus stops suddenly
5. _____ When you jump, the ground pushes up on you
6. _____ A bigger kick makes the ball fly faster

ACTIVITY 3: Forces & FBDs

1. In a balanced FBD, the net force = _____ and the acceleration = _____.
2. Which has MORE inertia: a basketball or a bowling ball? _____
3. Force is measured in _____ (unit).
4. If a car pushes 40 N right and friction pulls 15 N left, what is the net force? _____ N _____
5. A full shopping cart is _____ (easier / harder) to push than an empty one because of more _____.

I Can... (Self-Check for Milestones)

- | | |
|---|--|
| <ul style="list-style-type: none"> ☐ Define distance and displacement (S8P3a) ☐ Distinguish speed from velocity ☐ Define acceleration (including direction changes) ☐ Read a distance-time graph (slope = speed) ☐ Read a speed-time graph (slope = acceleration) ☐ Identify balanced vs. unbalanced forces (S8P3b) | <ul style="list-style-type: none"> ☐ Calculate net force from multiple forces ☐ State Newton's 1st Law (Inertia) ☐ Apply Newton's 2nd Law ($F = m \times a$) ☐ Identify Newton's 3rd Law (action-reaction) ☐ Explain how mass affects force needed (S8P3c) ☐ Read and draw free body diagrams |
|---|--|